

Automating water quality monitoring in aquaculture ponds: drone-based evidence from Tamil Nadu, India

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This study aimed to determine the effectiveness of drone-based autonomous systems, in combination with autonomous surface vehicles, autonomous underwater vehicles, Internet of Things systems and artificial intelligence applications, in real-time water quality monitoring. The study was performed over six months using experimental and control aquaculture ponds at the Aquaculture Research Facility in Pollachi, Tamil Nadu, India. Water quality parameters (dissolved oxygen, pH, turbidity, temperature, ammonia and nitrates) estimated using autonomous systems were compared to those monitored via conventional laboratory-based techniques. Machine learning algorithms were designed to use the assimilated data to identify and predict future anomalies and intervene. The average dissolved oxygen and pH in the experimental ponds (8.5 ± 0.7 mg l⁻¹ and 7.5 ± 0.2 , respectively) was much higher than that in the control ponds (6.9 ± 1.1 mg l⁻¹ and 7.1 ± 0.4), and the turbidity was lower in the experimental ponds (80 ± 10 NTU) than in the control ponds (95 ± 15 NTU). After intervention, the dissolved oxygen levels in the hypoxic areas increased from 5.5 to 7.8 mg l⁻¹ following aeration, indicating a 20–25% increase in 30 min, while the level of ammonia decreased from 0.9 to 1.2 mg l⁻¹ in 24 h, indicating that the bio-remediation agents were effective. These improvements translated into an increase in fish survivability (95%) and biomass gain (25%). These outcomes show that autonomous monitoring with the help of autonomous vehicles and artificial intelligence-based decision support can help to ensure timely intervention, enhance resource utilisation and increase sustainability in aquaculture.

Keywords: autonomous surface vehicles, autonomous underwater vehicles, Internet of Things, IoT, dissolved oxygen prediction, precision fish farming, smart aquaculture systems

Introduction

The use of drone-assisted autonomous systems for real-time water quality monitoring on aquaculture farms could be a major step forward in aquaculture operations. Conventional approaches to water quality assessment are costly in terms of time and effort, and may not offer contemporaneous results, which are critical to sustaining appropriate environments for aquatic fauna. Self-sufficient systems, especially those using drones, present a revolutionary approach to monitoring water quality; they can assess problems within the farming environment in a timely manner and boost production.

Recent studies have shown that autonomous surface vehicles (ASVs) are useful for environmental monitoring. Moreover, autonomous drone-based solutions with ASVs, autonomous underwater vehicles (AUVs) and Internet of Things (IoT) systems have proven useful in real-time water quality, dissolved oxygen (DO) and pH regulation and aquaculture growth (De Petris et al. 2025). For example, a self-propelled underwater vehicle was designed to search for efficient routes for monitoring water quality parameters, reducing human interference (Mohd Jais et al. 2024), and an ASV had sensors to record DO, pH and temperature and analyse these data in real-time (Liu et al. 2024). ASVs can be linked

with machine learning (ML) models, such as neural network and Random Forest models, to ensure that the data processing, validation and testing, and the performance measures (accuracy, RMSE and R²) are adequately reported, enhancing the reliability of such predictions. The use of AUVs in fish farms has also been investigated. AUVs have been used to monitor biofouling and the integrity of bon net enclosures in aquaculture (Rejeb et al. 2025). Furthermore, several studies have used neurocomputing techniques for the identification of toxic algal blooms, including a vision-based online algae monitoring system (Olanubi et al. 2024).

Aquaculture has grown rapidly due to increasing market demands for seafood and food security, alongside the increasing rates of natural resource consumption. However, water quality management, encompassing the health and productivity of aquatic life, remains complex. DO, temperature, pH and turbidity, as well as ammonia and nitrate levels, are crucial for appropriate growth conditions (Hemal et al. 2024). Previous techniques for monitoring these parameters, for example, those using vials and laboratory analysis, are slow and do not allow the user to handle problems in real-time. Free-roaming, self-organising drone-assisted systems offer a paradigmatic approach

to instantaneously and accurately monitor water quality in real-time (Liu et al. 2024).

Remote-operated aerial vehicles with enhanced sensor-endowed instruments and imaging capabilities have shown promise for environmental oversight. For example, systems based on unmanned aerial vehicles (UAVs) have been effectively used for the early observation of high-risk algal blooms in freshwater and marine ecosystems, and to monitor changes in chlorophyll and turbidity data with accuracy and good spatial coverage (Chen et al. 2024). Similarly, Diamante (2025) proposed a real-time UAV for monitoring aquaculture ponds; assessing the precision of DO and temperature measurements. These advances demonstrate the ability of drones to acquire dynamic water quality parameters across the large areas of aquaculture units.

ML and artificial intelligence (AI) algorithms improve the usability of drone-assisted systems. AI-based models can analyse water quality data and identify issues as they occur, predict future issues before they occur and even prevent issues before they occur (Lv et al. 2025). For example, Ganie et al. (2025) developed an AI-based technology to measure water quality (specifically DO and pH) in aquafarming, which performs more than 90% of the correspondence. Thus, the integration of AI with self-driving drones can greatly enhance aquaculture oversight and preventative responses.

As well as monitoring specific networks, autonomous systems are also being used for intervention and remediation. For localised water quality problems, drone delivery of aerators and bio-remediation agents to targeted areas in the aquaculture ponds has proven practical (Olivetti et al. 2023). Additionally, drones fitted with hyperspectral cameras have been applied for early indications of fish stress, achieving reduced mortality (Flores-Iwasaki et al. 2025). Compared to aerial drone systems, AUVs offer a similar but separate, and perhaps even more unique, perspective on water quality. AUVs equipped with multispectral sensors can gather comparative parameter data, including ammonia, nitrate and salinity levels, just below the water surface (Byrd et al. 2025). For instance, Aiyelari and Singh (2024) built an AUV system for aquaculture ponds that was able to survey DO distribution and localise low oxygen concentrations. Aquaculture water quality monitoring using aerial drones and AUVs is a systematic approach to data collection.

Both energy efficiency and sustainability should be central concerns when designing application-level autonomous systems for aquaculture. Concerning energy needs for continuous operation, solar-operated drones and AUVs have been tested. Rahul et al. (2024) implemented an extended-period solar energy-charged AUV system with low operational and environmental concerns. Likewise, an investigation into hybrid drones with photovoltaic systems revealed longer flight times that allowed longer inspection missions (Bilyeu et al. 2022).

Reducing the current variability in data acquisition, pre-processing and analysis methods is crucial for enabling the large-scale application of autonomous systems in aquaculture. Technologies such as cloud computing and IoT for drones and AUVs form efficient architectures for integrating and managing the big data collected by such

vehicles (Silvarrey Barruffa et al. 2021). Using an IoT-based approach reduces the cost of feeding aerators and other structures, as they can be controlled depending on water quality data collected from the aquaculture farm (Peck et al. 2024).

However, several challenges continue to hinder the utilisation of drone-assisted systems in aquaculture, including high start-up expenses and the demand for constant maintenance and skills that may impede operation, especially for small businesses (Pillay et al. 2024). In addition, more appropriate rules and regulations for drone application in aquaculture are required, owing to safety concerns and legal constraints (Ryu 2022). Mitigating these issues through low-cost designs, ease of interaction with drones and supportive policies is imperative for actualising the potential of drone-assisted systems in the aquaculture industry.

This research aims to:

1. Identify how an autonomous drone-assisted system can support constant, real-time water quality monitoring in aquaculture farms, and compare this with conventional manual monitoring techniques;
2. Assess the effectiveness of autonomous drones, ASVs, AUVs and other smart vehicles, equipped with state-of-the-art sensors, in measuring DO, pH, temperature, turbidity and ammonia and nitrate levels;
3. Determine the relevance of AI and ML algorithms in autonomous system predictive modelling, data anomaly detection and aquaculture;
4. Assess practical interventions and remediation through the delivery of aerators and bio-remediation agents to address localised water quality problems;
5. Ascertain the effectiveness of advancements in solar-hybrid configurations to improve the energy efficiency of these drone-integrated systems;
6. Assess the challenges of implementing such automated systems, focusing on time, labour, real-time monitoring, initial expenses, technical skills and legal restrictions, together with possible solutions including cost-efficiency designs, IoT-based approaches and extensive protocols for data control.

Materials and methods

Figure 1 shows the overall structure of the experimental design in terms of drones, ASVs, AUVs, IoT integration and intervention modules.

Study area and aquaculture setting

The study was undertaken at the Aquaculture Research Station, Pollachi, Tamil Nadu, located in South India, which features adjustable-scale ponds appropriate for experimental purposes. The facility has 10 rectangular ponds, 20 × 50 m in size with an average depth of 2.5 m, and facilitates aquacultural enterprises. Certain ponds were run as conventional ponds, while others were used as experimental ponds, receiving autonomous system treatment in addition to the other practices applicable to the control ponds. The selected fish species included tilapia (*Oreochromis niloticus*) and catfish (*Clarias gariepinus*), which are favoured in aquaculture owing to their market value and are hardy species that can survive in

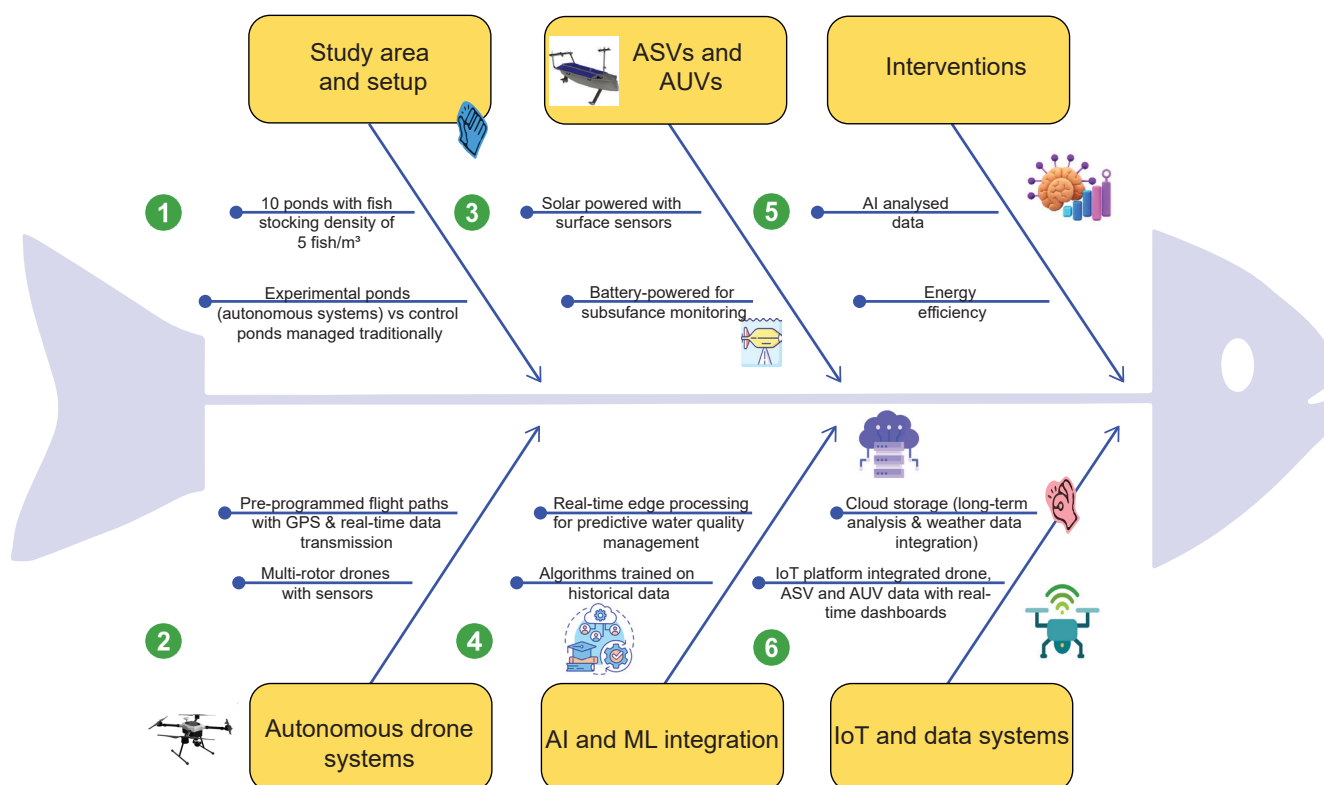


Figure 1: Schematic representation of the experimental setup showing the drones, autonomous surface vehicles (ASVs), autonomous underwater vehicles (AUVs), Internet of Things (IoT) platform and intervention mechanisms used in the aquaculture ponds

different water quality regimes. The stocking density was consistently limited to 5 fish/m³ to provide a realistic model of aquaculture conditions. In all ponds, the water quality was controlled with paddlewheel aerators and the water was changed whenever necessary. Within the peripheral area, mesophytic vegetation was maintained, and the matter accumulated by the water embankments also tended to provide an open control exposure [not sure what your intended meaning is here. Please try to clarify this sentence]. Further details about the environmental parameters and ponds are provided in Table 1.

Experimental design

The study employed a factorial experimental design, with three primary factors: (1) water quality levels (good, moderate and poor); (2) drone interventions for remediation (monitoring only, aeration and bio-remediation); and (3) time of day (morning, midday and evening). Treatments were applied with three repetitions, giving a total of 27 experimental units, every week for six months. The six-month timeframe was adopted to match seasonal alterations in water quality, as well as the growth performance of the fish. Individual and interactive effects were also considered, and a factorial design was used to test the effects of water quality conditions, as well as drone-based interventions and diurnal variability, providing strong statistical comparisons between the autonomous and conventional management strategies.

Autonomous drone systems

The UAVs conducted 450 monitoring flights to ascertain the quality of the water. The drone flight path was set to a grid with spiral flight patterns to ensure equal coverage of the ponds. The mapping exercise for each pond was completed within 30 min to capture variations in water quality.

Built-in and fully mobile multi-rotors were a key component of the monitoring system, introduced to provide tailored water quality readings. Flexible payload bay systems were provided on each of the drones, with options including innovative sensing technologies for parameters such as DO, pH, TSS, chlorophyll, temperature and nutrients such as ammonia, nitrate and phosphate. DO was measured using optical fluorescence sensors with high measuring accuracy (range: 0–20 mg l⁻¹), pH using solid-state glass electrode sensors which were adjusted once weekly for a reliable error of no more than ± 0.01 pH units, turbidity using nephelometric sensors that captured suspended-solid light scatter (range: 0–4 000 NTU) and temperature using a thermistor with an accuracy of ± 0.1 °C. To guarantee reproducibility, weekly calibration of all sensors was performed with certified standard solutions. The DO sensors were calibrated using zero-oxygen and air-saturated water and the pH sensors were calibrated using buffer solutions (pH 4.0, 7.0 and 10). Several coverage techniques, including grid and spiral flight patterns, were used by the drones depending on the dimensions of the ponds; all these flight paths used GPS-integrated mission planning

software. Using LoRaWAN for short-range transmission and 5G for long-range transmission allowed real-time data transmission to support data analysis. The sensors were calibrated every week using recognition standard solutions to avoid compass variability. Technical details of the drones are provided in Table 2.

The drones involved in the investigation were all quad-rotor UAVs (with a medium payload) and had onboard GNSS modules (horizontal error ≥ 0.5 m [Do you mean ≤ 0.5 m?]). The DO, pH, temperature and turbidity sensor sampling frequency was set at 1 Hz. DO was measured with a margin of uncertainty of ± 0.2 mg l⁻¹, pH had a margin of 0.01, temperature had a margin of 0.1 °C and turbidity had a margin of 2% of the full scale. To initiate re-calibration, sensor variance was applied based on laboratory cross-validation against certified reference standards, during which turbidity and DO sensors had to be brought within $\pm 3\%$ of the full-scale range, pH sensors within ± 0.05 and temperature sensors within ± 0.2 °C of the range.

To prevent sensor fouling owing to biofilm growth and suspended sediments, mechanical wipers and anti-fouling protective caps were fitted on all the submerged sensors. The sensors were also wiped with non-abrasive, softened brushes and distilled water every two weeks. This minimised signal drift and signal noise during long-term deployment.

Design and development of autonomous vehicles

We also relied on ASVs and AUVs to complement the drones in conducting comprehensive water quality surveys. The ASVs measured epiphenomena at the surface of the water column, including temperature gradients and nutrient concentrations. They were solar-powered and carried surface-level measurement tools, including thermometers for temperature gradients and algae density, light meters for turbidity and density meters for nutrients. The AUVs surveyed the vertical distribution of water quality and were used to gather information on DO stratification and the ammonia and nitrate content of the sediments. These vehicles moved independently using sonar commands and inertial platform control, thereby maintaining their position and avoiding obstacles. While the ASVs were charged by solar panels with a 50 W capacity, the AUVs were equipped with high-capacity lithium-ion batteries that allowed them to operate for 10 h continuously. Details of these vehicles are presented in Table 3.

The speed of the ASVs corresponded to pre-programmed GPS transects with 5 m intervals between waypoints. The AUVs followed depth-layered profiles at depth intervals of 0.5 m. The positioning error of the ASVs was ± 0.3 m, whereas that of the AUVs was ± 0.5 m. The underwater sensors were pressure-compensated and recalibrated every two weeks to compensate for sediment fouling.

Artificial intelligence and machine learning

A supervised ML framework was used to predict the water quality and identify anomalies. Separate regression equations were used to predict the DO, pH and turbidity so that each parameter could perform optimally. The neural

network architecture was the same for all parameters, although each parameter was trained by sampling that independent parameter, with the respective target variables. A feedforward multilayer perceptron neural network was used for regression assignments (DO, pH and turbidity) and comprised the following layers:

- Input layer: seven features (DO, pH, temperature, turbidity, ammonia, feeding time and time of day)
- Hidden layer 1: 64 neurons with rectified linear unit (ReLU) activation
- Hidden layer 2: 32 neurons with ReLU activation
- Output layer: 1 neuron with linear activation
- Hyperparameters included:
 - Optimiser: Adam
 - Learning rate: 0.001
 - Loss function: mean squared error
 - Batch size: 32
 - Epochs: 150
 - Early-stopping patience: 15 epochs

As a classifier, a Random Forest of 100 trees and a maximum depth of 10 was trained using Gini impurity to identify anomalies (hypoxic events and nutrient overload). Anomaly detection applications were the only tasks that were classified using metrics (precision, recall, F1-score).

The data were separated into training (70%), validation (15%) and testing (15%) datasets. The model was a neural network with two hidden layers (with durations of 64 and 32 neurons) and ReLU. To be able to use continuous-error assessment, root mean square error (RMSE) and R^2 were utilised to assess the models used in regression activities.

The models were coded in Python, as well as the statistical learning libraries TensorFlow and Scikit-learn. Smart devices at the site facilitated data analysis in near-real time, making it easier to respond to hypoxic zones with aerators. To reduce fish stress and death, predictive analytics were used through set benchmarks, key performance indicators and alerts.

The regression tasks were the prediction of DO, pH and turbidity. Hence, model performance was assessed in terms of the continuous error (RMSE and R^2). The classification-based metrics, such as precision, recall and the F1-score, were used exclusively for anomaly finding, such as detecting cases of hypoxia or nutrient overload.

Drone application

The drones were fitted with provisions for select operations. Following AI-based predictions based on real-time data, drone-based interventions were intentionally implemented at the sites requiring the most remediation, according to a standardised protocol: (1) prediction of hypoxic or nutrient-enriched regions by AI; (2) autonomous retrieval of drones to a known location; (3) deployment of aerators through a release mechanism or spraying bio-remediation agents; and (4) intervention monitoring.

Aerators were connected to an adaptive mechanical system for rapid deployment when depleted oxygen regions were identified during monitoring. The aerators were small floating diffused-air aerators (equivalent oxygen transfer capacity: 0.5 HP) with micro-bubble diffusers. The units were placed in the pond in advance and activated by drone-controlled wireless relay rather than being literally

Table 1: Environmental parameters

Parameter	Value	Methodology
Pond dimensions	20 × 50 m	Direct measurement
Average depth	2.5 m	Depth gauge
Stocking density	5 fish/m ³	Counting and weighing
Water source	Borewell and rainwater	Water source analysis
Aeration system	Paddlewheel aerators	Visual inspection
Species cultured	Tilapia, catfish	Farm stocking records

Table 2: Technical details of drones

Feature	Specification	Purpose
Flight time	30 min	Extended monitoring
Payload capacity	2 kg	Sensor and intervention modules
Navigation system	GPS-enabled	Precise route mapping
Communication protocol	LoRaWAN, 5G	Real-time data transmission

Table 3: Vehicle types and their power source, navigation system and monitoring parameters

Vehicle Type	Power Source	Navigation System	Monitoring Parameters
ASV	Solar + lithium battery	GPS and inertial	Temperature, turbidity, ammonia and nitrates
AUV	Lithium-ion battery	Sonar and inertial	DO, sediment ammonia and nitrates

carried and deployed by the drones. In the control ponds, the paddlewheel aerators were not turned off for comparison.

In areas with concentrated nutrients, bio-remediation agents were delivered through discrete, calibrated spraying modules. Nitrifying bacterial consortia containing *Nitrosomonas* spp. ($\geq 10^8$ CFU ml⁻¹) and *Nitrobacter* spp. ($\geq 10^8$ CFU m⁻¹) were used as bio-remediation agents with a dosage of 0.5 L ha⁻¹, representing a relatively high active bacterial concentration of about 5×10^7 CFU m⁻² of the treated water surface. The effectiveness of these measures was assessed by collecting post-intervention data on water quality 30 min, 6 h and 24 h after intervention to determine recovery.

Energy efficiency and sustainable energy

The sustainability and energy efficiency of the autonomous systems was investigated. All ASVs had a solar power system that could generate up to 50 W under good sun conditions to minimise use of the external power supply. Some of the unmanned aircraft used lithium-ion batteries with regulated charging cycles. A comparative evaluation of the energy efficiency under various conditions of operation is provided in Table 4.

Data acquisition and IoT incorporation

A web-based IoT platform helped to enable the DA&AS<What is DA&AS? data acquisition and analysis OR drone-assisted autonomous systems? This abbreviation is not used anywhere else. Please provide term>. Using LoRaWAN and 5G, the drones, ASVs and AUVs gathered data and shared it with a central, integrated and long-term cloud data storage suite in real-time. The platform also featured weather data integration to contextualise the other data and improve predictions.

Table 4: Energy efficiency under various operating conditions

Metric	Drone System	ASV System
Battery capacity (mAh)	10 000	25 000
Solar panel output (W)	N/A	50
Average runtime (h)	4	8
Charging time (h)	2	6

System reliability and fail-safe mechanisms

Because the architecture was multi-layered and included drones, ASVs, AUVs and IoT integration, redundancy and fault tolerance were added to avoid the complete collapse of the entire system in the event of one subsystem malfunctioning, so as not to affect monitoring. The system architecture was developed based on a modular system, in which the individual monitoring platforms (drones, ASVs, AUVs) can run independently but share data with the central IoT cloud server. If one of the subsystems was not available, another monitoring technology could be used as a reserve. For example, in the event of drone failure, surface monitoring could be maintained with the help of ASVs and vertical profiling could be maintained with the help of AUVs; and should an AUV sensor fail, the surface and drone measurements could compensate. In the event of communication failure, the smartphone units temporarily stored the information (up to 48 h) and reconnected as soon as a connection was re-established. To alert operators to fault or failure, an automated alert system was implemented if sensor values were outside of anticipated physical limits, more than 15 min of communication was intercepted or battery levels fell below 20%.

No complete system shutdown occurred withing the six-month experiment. Two standalone sensor interruptions were identified (AUV turbidity probes) and corrected within 24 h without compromising data continuity.

Water quality assessment

During the field tests, water samples were collected manually and periodically to confirm the sensor data. Conventional laboratory techniques were used for analyses, such as DO determination by Winkler's titration, pH determination using glass electrodes and ammonia and nitrate analysis via spectrophotometry.

Statistical analysis

The collected data were analysed in R and SPSS software. To perform the analysis and identify the main and interaction effects of the drone interventions and environmental conditions on the multiple dependent variables of water quality, a two-way between-groups ANOVA was used. Tukey HSD post hoc analysis was used to determine group differences. The validity of the assertions was determined by Pearson's correlation coefficients, which compared the autonomous system data with the manual measurements.

Results

Study area and aquaculture setup

The field study, performed at the Aquaculture Research Facility in Pollachi, Tamil Nadu, established the effectiveness of the introduced autonomous systems and demonstrated improvements in overall water quality and indicators of fish health in the experimental ponds. During the six-month study, the water quality conditions in the experimental ponds were relatively stable, constant and closer to ideal than those in the control ponds, where fluctuations and occasional abnormalities were observed. DO levels, which were corrected in the experimental ponds, were significantly higher ($8.5 \pm 0.7 \text{ mg l}^{-1}$) than those in the control ponds ($6.9 \pm 1.1 \text{ mg l}^{-1}$; $p < 0.01$), and a pH of 7.5 was observed in the experimental ponds. The water clarity was improved

by the self-organised solutions in the experimental ponds, with the turbidity decreasing by 16%. Tukey's HSD test confirmed statistically significant overall variation between the experimental and control ponds ($p < 0.05$).

Similar trends were observed in fish performance metrics. The survival rate in the test culture ponds was 95% while that in the control ponds was only 85%. The growth performance enhancement, in terms of weight gain and the feed conversion ratio, observed in the experimental ponds was beneficial and meaningful. **In individual ponds** [It is unclear what you mean by "In individual ponds" here. Are these for just two individual ponds (one experimental and one control) or were all the fish in experimental/control ponds pooled for a single value? There is no error reported in the table for the fish metrics, unlike the water quality parameters. Please clarify this both in the text and the Table], the mean weight gain was 250 g/fish in the environmentally challenged experimental treatment ponds compared to 200 g/fish in the control ponds ($p < 0.05$). The relationships between water quality and fish health parameters are presented in Table 5.

Autonomous drone systems

The findings indicate that the water quality parameters were spatially homogeneous. This highlights the issue of spatial variation, and hence the need to intervene in specific areas. A representative image of the spatial distribution of DO levels obtained from drone sensors is presented in Figure 2.

Validation of the drone-collected data against manual measurements revealed strong correlations (Table 6). The correlation coefficients for DO, pH and turbidity were 0.98, 0.97 and 0.96, respectively. Despite these high correlations

Table 5: Comparison of water quality and fish health metrics

Parameter	Experimental Ponds (Mean $\pm$ SD)	Control Ponds (Mean $\pm$ SD)
Dissolved oxygen (mg l^{-1})	8.5 ± 0.7	6.9 ± 1.1
pH	7.5 ± 0.2	7.1 ± 0.4
Turbidity (NTU)	80 ± 10	95 ± 15
Survival rate (%)	95	85
Weight gain (g/fish)	250	200
Feed conversion ratio	1.3	1.5

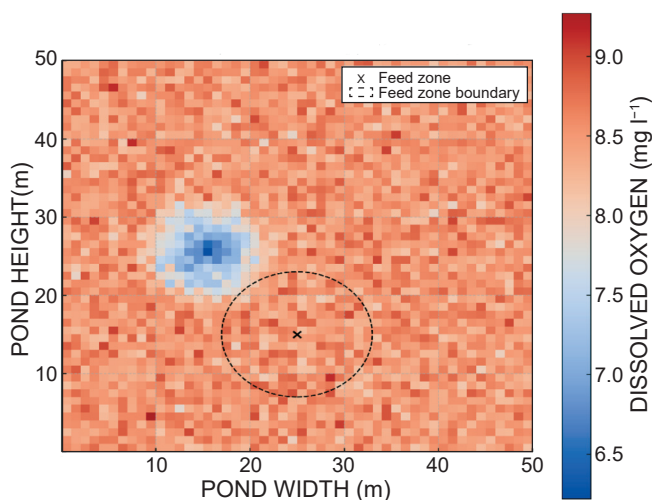


Figure 2: Heatmap of dissolved oxygen levels across an experimental pond

Table 6: Coefficients of determination between the sensor-derived and manual measurements

Parameter	Sensor Data Range	Manual Data Range	Correlation Coefficient (r)
Dissolved oxygen (mg l^{-1})	6.5–10.2	6.4–10.0	0.98
pH	7.0–8.5	6.9–8.4	0.97
Turbidity (NTU)	50–120	48–115	0.96

Table 7: Subsurface water quality parameters recorded by autonomous underwater vehicles

Depth (m)	Dissolved oxygen (mg l ⁻¹ , Mean ± SD)	Temperature (°C)	Turbidity (NTU)	Ammonia (mg l ⁻¹)	Nitrate (mg l ⁻¹)
Surface (0.5)	8.8 ± 0.5	28.5 ± 0.3	75 ± 10	0.5	1.2
Bottom (2.0)	6.5 ± 0.8	27.0 ± 0.5	90 ± 12	1.5	3.2

Table 8: Energy requirement comparison of drone and autonomous surface vehicle systems

Metric	Drone System	ASV System
Battery Utilisation (%)	90	85
Solar Contribution (%)	N/A	40
Average Runtime (h)	4	8

between the sensor and manual measurements, sensor drift was detected during the six-month deployment. Calibration was thus maintained weekly and laboratory validation was conducted periodically to prevent long-term drift and guarantee the reliability of the data across the entire monitoring period.

Design and development of autonomous vehicles

Strong vertical stratification was observed using the AUVs. For instance, DO levels dropped from 8.8 ± 0.5 mg l⁻¹ at the surface of the water column to 6.5 ± 0.8 mg l⁻¹ at a water depth of 2 m (Table 7), while the turbidity and ammonia and nitrate concentrations were higher at the bottom of the pond in the proximity of the sediment layer.

Artificial intelligence and machine learning integration

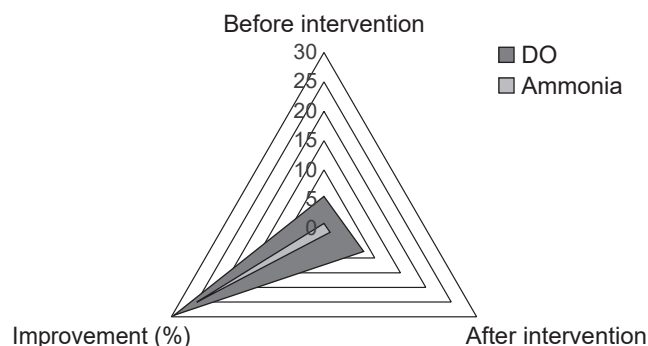
When the prediction software results were compared with test data for DO, the RMSE of the neural network was 0.62 mg l⁻¹ and the coefficient of determination (R^2) was 0.91. The pH prediction model gave RMSE = 0.18 and R^2 = 0.88 and the turbidity prediction model gave RMSE = 7.6 NTU and R^2 = 0.87. These findings demonstrate the good predictive ability of the regression model for water quality estimation.

Practical interventions of drones

The drones performed 20 targeted interventions throughout the study, using aerators twice as often as bio-remediation agents. Figure 3 shows the effect of drone-assisted interventions on water quality. This study established that water aeration deployments raised the DO levels in hypoxic areas by 20–25% within half an hour. For instance, the DO levels in a zone with an initial reading of 5.5 mg l⁻¹ increased to 7.8 mg l⁻¹ after aeration. Bio-remediation agents were effective in decreasing ammonia levels from 1.2 to 0.9 mg l⁻¹ within 24 h. As Figure 3 shows, directed aeration rapidly restored DO levels (within 30 min) and bio-remediation led to some ammonia reduction (within 24 h), indicating the success of the spatially specific interventions.

Energy efficiency and sustainability

Energy is crucial for any autonomous system, and the drones achieved 90% battery utilisation efficiency, flying

**Figure 3:** Quantitative impact of drone-guided interventions on water quality parameters. Improvement (%) indicates the relative change in the concentration of dissolved oxygen and decrease in the ammonia concentration with reference to the values before intervention

for an average of 27 min on a single charge. The ASVs, powered by 50 W solar panels, ran for an average of 7.8 h per day, using approximately 40% solar energy. Table 8 summarises the energy characteristics of the drones and ASVs.

Data acquisition and IoT integration

The employed IoT platform received and analysed over 500 GB of traffic data from drones, ASVs and AUVs. Real-time dashboards gave a continuous overview and instant notification of events. Successful integration of external weather data enhanced the predictive capability of the model during rain events, which caused turbidity changes, as illustrated in Figure 4.

Drone intervention vs environmental influence

By comparing quantitative data, it was also possible to establish which systems were effective in monitoring water quality. Drone intervention and environmental influence showed statistically significant effects on both DO levels ($F_{2,24} = 18.6$, $p < 0.01$) and water turbidity measurements ($F_{2,24} = 9.4$, $p < 0.05$), according to two-way ANOVA (Table 9). Pearson correlation coefficients confirmed the accuracy of the sensors, with values > 0.95 for all parameters.

Discussion

The combination of AUVs and ASVs with autonomous drones has been instrumental in enhancing the real-time monitoring of water quality in aquaculture. For example, the use of drones has been adopted in the Andhra Pradesh region in India to monitor and control

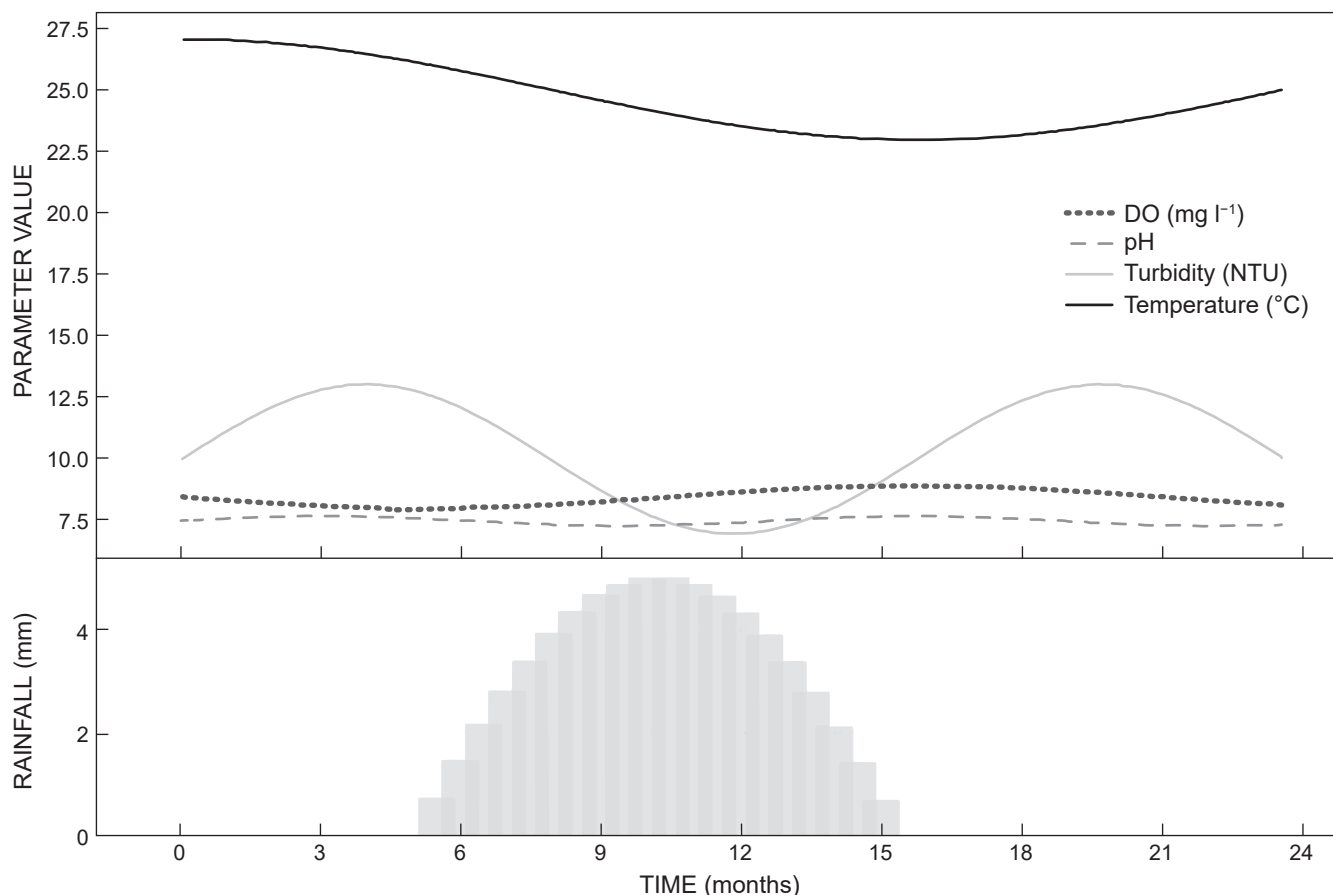


Figure 4: Internet of Things dashboard displaying real-time water quality trends

Table 9: Two-way ANOVA results for water quality parameters

Parameter	Source	df	F-value	p-value
Dissolved oxygen	Drone intervention	2.24	18.6	<0.01
Dissolved oxygen	Environmental influence	2.24	12.4	<0.01
Turbidity	Drone intervention	2.24	9.4	<0.05
Turbidity	Environmental influence	2.24	7.1	<0.05

the difficult water quality challenges on shrimp farms (Sundararajan et al. 2025). By enhancing constant pH and DO monitoring, these systems have made it easier for farmers to decrease incidences of shrimp mortality by up to 30% (Sundararajan et al. 2025). In the same way, on Norwegian salmon farms, variable ASVs with multi-parameter sensors have been used to monitor oxygen distribution and nutrient concentrations, and as a result, feeding rates were optimised and waste production minimised (Hridoy et al. 2025).

In China, drone components and AI models have been used in carp farming to predict and detect toxic and adverse water conditions, reaching about 95% accuracy and enhancing productivity by 25% (REFERENCE REQUIRED). Thus, the effectiveness of these systems can improve the accuracy and scope of the management of important parameters, including DO, pH, temperature, turbidity

and ammonia and nitrate concentrations, as part of their high-performance profile. Moreover, they are versatile in most aquaculture environments, and are therefore considered indispensable instruments for sustainable aquaculture. Some advancements were developed more recently, including LoRaWAN and 5G communication protocols, which transmit data in real-time to improve operation (Tran et al. 2024; Xu et al. 2025).

These sophisticated autonomous vehicles with sensing technology are a less expensive approach and provide real-time results, as opposed to manual sampling and laboratory analysis which are time consuming and costly (Pillay et al. 2024). Some autonomous vehicles have optical fluorescence sensors that allow for precise and stable DO and turbidity measurement, removing the influence of the chemical reagents used in manual titration approaches and reducing material costs by 40% (Sutrisno and Rahmat

2025). Nephelometric turbidity sensors allow real-time monitoring, unlike the gravimetric method, rendering nephelometric turbidity sensors over 50% more effective in large-scale aquaculture (Bouzidi et al. 2025). Such strides have allowed the development of enhanced monitoring systems for aquaculture farms that are efficient, affordable and easily deployable. Further technological advancements include self-driving vehicles that use sonar sensors and inertial navigation systems to categorise the vertical stratification of ammonia and nitrate concentrations down to the sediment (Soares et al. 2024; Tina et al. 2025).

Drones can be used to oxygenate hypoxic zones and restore nutrient concentrations through bio-remediation agents. During our study, targeted aeration increased the DO in hypoxic zones by 25% and was crucial for avoiding fish stress and death. Similar findings were observed by Capetillo-Contreras et al. (2024), who reported that the use of drone-based feeding resulted in similar enhancements in water quality in tilapia farms. In confined experimental ponds equipped with these systems, timed aeration for DO enhancement removed fish stress in hypoxic areas and increased DO concentrations from 5.5 to 7.8 mg l⁻¹ within 30 min following targeted aeration (Capetillo-Contreras et al. 2024). Bio-remediation interventions resulted in a 25 % decrease in ammonia levels, from 1.2 to 0.9 mg l⁻¹ after 24 h, enhancing water quality (Capetillo-Contreras et al. 2024). In addition, through hyperspectral imaging, harmful algal blooms were detected early; the frequency of dangerous blooms was minimised by 15% compared to conventional methods. This emphasises the efficiency of drone assistance in overcoming certain significant obstacles in aquaculture, allowing timely intervention and enhancing the productiveness of monitoring systems. For practical remediation, these systems carry aerators or calibrated nutrient sprayers. Moreover, by combining drones with hyperspectral imaging and AI algorithms, the early detection of harmful algal blooms and fish stress was improved. Artificial neural networks and random forest classification models offered over 90% predictive accuracy for adverse conditions, enabling timely intervention (Nagothu et al. 2025; Yang et al. 2024).

Autonomous solar vehicles and hybrid aerial systems with solar panels improve sustainability. Solar contributions alleviated dependence on conventional power sources, increasing operational hours and improving the flight time of drones. When equipped with sensors, these ASVs also have better energy characteristics than other aeration and water control mechanisms in aquaculture. For instance, ASVs used in solar energy tests ran an average of 8 h daily and used 40% solar energy, greatly reducing power costs (Soon et al. 2025). Meanwhile, conventional aeration systems require 30% more power to provide similar operation times (Eze et al. 2023). ASVs with solar panels also have longer flight times, as they are less reliant on batteries. Their use represents a 25% reduction in carbon emissions on average when compared to traditional drones with hybrid photovoltaic systems (Soon et al. 2025). Such advancements leave no doubt that modern autonomous systems have many advantages, further enhanced by modern techniques, in reducing both costs and environmental impacts.

While there are advantages to using autonomous systems, difficulties in expanding such systems persist. In the United States, the Federal Aviation Authority regulates the operation of drones and their airspace compliance, and can thus limit their wide application on aquaculture farms (Arabelli et al. 2025). Similarly, in the European Union, drone operation has become challenging due to General Data Protection Regulation compliance regarding the processing of data collected in real-time by these systems (Arias et al. 2025). Smallholder farmers in India also face legal challenges, including the lack of a standard regulating policy for the use of drones by the Indian government and high import taxes on drone technology (Tran et al. 2024). To realise the full potential of autonomous systems in aquaculture, these systematic regional differences in regulation must be tackled through policy reforms and global partnerships. Possible problems include capital, powering and maintenance costs and the technical ability required to manage the system, which can make such systems extremely expensive and unaffordable for small-scale aquaculture farmers (Fonseca et al. 2023; Pillay et al. 2024). Moreover, there are some legal limitations related to the use of drones in agriculture and aquaculture production. These problems are best solved through affordable designs, simple interfaces and favourable legal environments to expand the market.

A cost-benefit analysis was performed on the proposed monitoring system using autonomous vehicles to determine whether it is economically feasible. The capital expenditure to install a complete system with one drone unit (with multi-parameter sensors), one ASV, one AUV and IoT infrastructure fell between INR 7.5–8.5 lakhs according to the setup. Although this could be a heavy investment for smallholder farmers, the following three deployment models could enhance affordability:

1. Cluster-based cooperative ownership model: a single system held between 5–10 farmers in a radius of 5 km.
2. Service-based model: monitoring offered as a subscription (e.g. INR 8 000–12 000 per month).
3. Phased implementation model: initial drone and IoT platform implementation only at INR ~4.5 lakhs, followed by ASV/AUV integration, if required.

The benefits included a 10% higher survival rate, a 25% biomass increase and 40% less manual labour in monitoring. In the semi-intensive pond system with a 1 ha area in Tamil Nadu, this equates to an estimated additional revenue of INR 2.0–2.5 lakhs per production cycle. In the case of cooperative deployment, the projected payback period would be reduced to 1.5 years. Therefore, full deployment can be capital-intensive, but with the help of shared and scalable operational models, the system is economically viable in medium-sized and cooperative aquaculture farms.

IoT platforms and cloud computing have boosted the effectiveness of large-data strategies and predictive analytics. The characteristics of the AI and ML algorithms can also be further researched to enhance the predictive accuracy and operation of a system. Moreover, the emergence of self-controlling systems with learning features should allow for automatic optimisation processes, minimising human interference (De Petris et al. 2025).

From an environmental perspective, the implementation of best management practices for monitoring and managing water quality can minimise the impacts of aquaculture activities. By reducing nutrient input and avoiding toxic algal blooms, autonomous systems may make aquaculture less environmentally damaging (Eze et al. 2023).

Systematic application of the practical interventions occurred, including drone-deployed aerators and calibrated bio-remediation agents; after intervention, DO levels increased by 20–25% within 30 min and ammonia concentrations decreased by 25% within 24 h. Autonomous systems are not only justified in the aquaculture domain but are also congruent with other areas of geospatial AI and Earth Observation systems. Applications include monitoring infectious disease outbreaks (Orusa et al. 2024a; Viani et al. 2024), assessing municipal contributions to the environment through Sentinel imagery (Orusa and Borgogno Mondino 2021; Viani et al. 2023, 2024; Orusa et al. 2024b), predicting global water quality with Web-GIS (Viani et al. 2025) and monitoring environmental hazards to wildlife health with autonomous remote sensing, IoT and AI frameworks (Orusa et al. 2023a,b,c,d).

The suggested multi-platform system must not be viewed as an isolated development, as more general progress in Earth Observation-based environmental monitoring systems is also pertinent. In contemporary research, geospatial analytics and satellite data have been used to undertake large-scale environmental risk assessments, such as wildlife disease monitoring (Orusa et al. 2020, 2025a), climate-driven vegetation dynamics (Orusa and Borgogno Mondino 2021) and thermal risk mapping via Landsat time series (Viani et al. 2023). Such studies have emphasised the growing importance of Earth Observation data in aiding evidence-based environmental management. Although these methods are usually used at a regional or landscape level, our analysis demonstrates an interoperative application at a micro-level, transferring Earth Observation principles to a monitoring framework with pond-level accuracy (Orusa et al. 2025b).

Similarly, the merging of UAV-based multispectral sensing to estimate water and wastewater parameters (Agrawal and Narulkar 2026) and hybrid IoT–drone–AI systems to manage water (Elkhodr & Al-Sabbagh 2026; Zhang et al. 2025) signals a declining separation between aerial robotics and water-sensing systems. This places the study within the paradigm shift towards decentralised, autonomous and data-driven environmental surveillance infrastructures that can be applied in agriculture, urban climate monitoring and ecosystem health management (Hu et al. 2022; Song and Yang 2023; Jiang et al. 2024; Sudharsan et al. 2024; Zhang et al. 2024; Zhou et al. 2024; Peng et al. 2025; Dai et al. 2026; Li et al. 2026).

Several design and policy changes are suggested to enhance regulatory compliance and adoption. These include the implementation of geofencing and altitude-constrained flight control to meet national aviation regulation requirements, on-gadget anonymity systems and domestic data storage to mitigate data privacy issues, and modular systems that allow incremental investment as opposed to full scale initial deployment. Subsidies on drones used in agriculture, cooperative forms of ownership

and training programs directed at farmers would reduce the barriers experienced by small fish farmers.

Conclusion

The integration of autonomous drone-assisted systems accompanied by ASVs, AUVs and IoT platforms has revolutionised aquaculture management by providing accurate, timely water quality monitoring and control. When used in experimental ponds, these technologies preserved optimal environmental conditions, improving fish health, survival and growth rates. The use of AI allowed predictive measures that addressed water quality issues before they actually occurred. Other cost savings and impacts included sustainable energy use and solar-powered ASVs. These are the main advantages of the use of autonomous systems over traditional aquaculture, which faces problems such as high labour intensity and low response speeds. The existing problems associated with autonomous systems, such as high initial costs and the need for qualified staff, remain relevant, but the specified technologies demonstrate their potential for improved performance and environmentally friendly use. Extending the implementation of such systems to a larger population will be essential for the advancement of precision aquaculture. Although the autonomous multi-vehicle system requires a large initial investment and periodic maintenance, the modular and redundant structure demonstrated operational stability, economical scalability and decreased reliance on a long-term human workforce under real aquaculture field conditions.

Limitations and practical challenges

Although the autonomous drone monitoring system has shown promising results, there are certain operational limitations that must be considered. Extreme weather conditions, such as strong winds and heavy rainfall, may disrupt the stability of drone flights, the acquisition of accurate data and the scheduling of monitoring missions, especially during monsoon seasons. Moreover, the extended use of water quality sensors can lead to fouling by biofilms and sediments, resulting in a loss of measurement accuracy, forcing the operator to maintain and recalibrate the sensor frequently. The initial cost of acquiring drones, autonomous vehicles, sensors and infrastructure to support the IoT can also be inhibitive, particularly for small-scale aquaculture farmers. Moreover, to ensure the successful functioning of these autonomous systems, skilled human resources with technical skills in drone handling, data interpretation and system maintenance are also needed. Overcoming these issues would involve enhancing sensor longevity, automatic calibration, economical system architectures and user-friendly interfaces. This would allow the large-scale, as well as long-term, implementation of such systems in aquaculture practices. Human maintenance was also operationally limiting as (1) the batteries needed to be inspected and replaced every two days and the sensors needed calibration once a week; (2) the solar panels on the ASVs needed cleaning every 10 days and the battery needed to be checked once a month; and (3) the sensors on the AUVs needed to be

cleaned every 14 days to prevent biofouling and the battery needed recharging after 8–10 h of operation. Biofouling was moderate in warmer seasons and occurred mostly on the turbidity and ammonia probes. Signal degradation was reduced by approximately 60% using anti-fouling coatings and mechanical wipers; nevertheless, **manual cleaning was required every 2 months [Above you say ‘the sensors on the AUVs needed to be cleaned every 14 days to prevent biofouling’ and here you say ‘manual cleaning was required every 2 months’. Please check this and clarify the difference if necessary]**. The average overall time of human intervention was 4–5 h/week for the entire system, which remains less than the average time required to manually collect samples and test them in a laboratory (12–15 h/week). The performance of the battery did not change during the study, and the lithium-ion system maintained > 90% capacity over 6 months. There were no significant cases of battery degradation.

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Ethics approval and consent to participate — No human participation was involved in this implementation process.

Human and animal rights — No violation of human or animal rights was involved.

Data availability statement — The raw datasets generated during this study comprise high-frequency, multi-sensor drone, ASV and AUV recordings exceeding 500 GB in size, system-specific calibration files, mission logs and operational telemetry data. Access is subject to institutional approval and data-use agreements to ensure responsible reuse. The following processed materials are available upon reasonable request: (1) aggregated weekly mean water quality datasets (dissolved oxygen, pH, turbidity, temperature and ammonia), (2) AI model architecture and training scripts (TensorFlow/Python), (3) hyperparameter configuration files, (4) intervention logs (aeration and bio-remediation events with timestamps) and (5) statistical analysis outputs (ANOVA tables and post-hoc test results).

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